$S = \{1, 2\}$

$$\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix} \times \begin{bmatrix}7 & 1\\ 2 & 8\\ 3 & 4\end{bmatrix}$$

$\det A_S \cdot \det B_S = (-3) \cdot (54) = -162$

$S = \{1, 3\}$

$$\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix} \times \begin{bmatrix}7 & 1\\ 2 & 8\\ 3 & 4\end{bmatrix}$$

$\det A_S \cdot \det B_S = (-6) \cdot (25) = -150$

$S = \{2, 3\}$

$$\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix} \times \begin{bmatrix}7 & 1\\ 2 & 8\\ 3 & 4\end{bmatrix}$$

$\det A_S \cdot \det B_S = (-3) \cdot (-16) = 48$

$\det(AB) = -162 - 150 + 48 = -264$: one product of minors per choice of columns